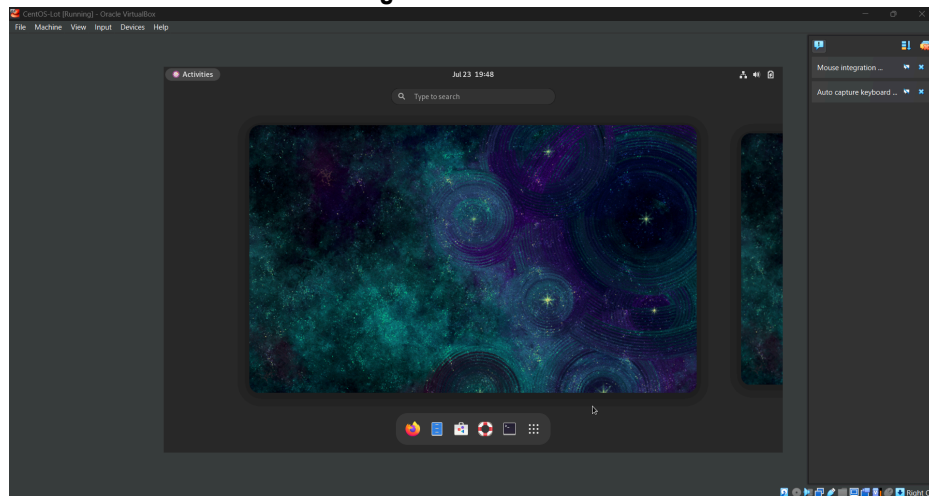


COURSE CODE/TITLE:	CPE 212	SECTION:	CPE31S2
DATE PERFORMED:	Jul 23, 2026	DATE SUBMITTED:	Jul 23, 2026
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ACTIVITY NO. 3.1 Install SSH server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image



Task 2: Install the SSH server package openssh

```

mdmlot@vbox:~$ dnf install openssh-server
Error: This command has to be run with superuser privileges (under the root user
on most systems).
[mdmlot@vbox ~]$ sudo dnf install openssh-server

We trust you have received the usual lecture from the local System
Administrator. It usually boils down to these three things:

#1) Respect the privacy of others.
#2) Think before you type.
#3) With great power comes great responsibility.

For security reasons, the password you type will not be visible.

[sudo] password for mdmlot:
CentOS Stream 9 - BaseOS                4.9 MB/s | 9.0 MB    00:01
CentOS Stream 9 - AppStream             14 MB/s | 28 MB     00:02
CentOS Stream 9 - Extras packages       16 kB/s | 21 kB     00:01
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[mdmlot@vbox ~]$

```

Task 3: Copy the Public Key to CentOS

```
mdmlot@vbox:~$ ifconfig lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 17 bytes 2045 (1.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 17 bytes 2045 (1.9 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[mdmlot@vbox ~]$ systemctl start sshd
[mdmlot@vbox ~]$ systemctl enable sshd
[mdmlot@vbox ~]$ cat ~/.ssh/authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQACxoZeJ5frCoibmMvEhLuk8DG6Ub4/5NB4wze0tzKaQ
lzmI5wyJKBzoc26vK5ahbvRASTd2YKLGKc2uhIUiy4+3fahpY8JpL/Amky7iTPo/+QCjXE7u3EvY/4cu
XzXR90NIyxceRGaCRyjMQcOK2Ydoyr8xjpmLKFRjohHo3HozhrBLqvp+4hd7IoJQEi6xkj37hfWroW/i
Ah2CTAL5689grhRoSGxWI3M9dP5tfofHTPbBS3Ytfk7gi/q4MpuDLSGcBE6pKkcyNlxXfT2I+FzIq0oH
COMdTPQ63ppH4V0XK1fWs+zkTKlQ68gz+/x0hCFPOM1AlZ6waHE3awNDyM8SIkH3PBWJ9tNFXyuEeKwF
hZHXX8rdV/E8Jot41+YRz6Qcnl7KGPx3/HSIyYM94VwHcTcmiWHhy+4Ktp9HtqumyCDPjjNVqS7yJBfy
H7sj4niP5ng24EVudRQITFSzS8MDcKDaYwEQM3/WEg4oUirIMBR9NW+Z9si46SNOtySi8nyQ4YNYLj6V
U01kQM5BRn0w7bjc30TBYfgpN9awqgYeixqQ6A1r6jThI6DcF03ICByXxSWD1Cfw0YWF0bG74WSwXufY
TD4+Ssuf0usMFgz0k2sfPZZ1DBEtUzCiIN66l7WNjFtAhw2MCVfoYJq4EHLfT9xRSDuR0wMAPyq2X/Nm
gw== mdmlot@mdmlot-VirtualBox
[mdmlot@vbox ~]$
```

Task 4: Verify ssh remote connection

```
mdmlot@vbox:~$ ssh mdmlot@192.168.56.103
ED25519 key fingerprint is SHA256:mQ2Tn8DAe44Fk0+g/oLb28sG/6+I3ln4QmFrKfRueU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
mdmlot@192.168.56.103's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh 'mdmlot@192.168.56.103'"
and check to make sure that only the key(s) you wanted were added.

mdmlot@mdmlot-VirtualBox:~$ cat ~/.ssh/authorized_keys
cat: ~/.ssh/authorized_keys: No such file or directory
mdmlot@mdmlot-VirtualBox:~$ cat ~/.ssh/authorized-keys
cat: ~/.ssh/authorized-keys: No such file or directory
mdmlot@mdmlot-VirtualBox:~$ ssh mdmlot@192.168.56.103
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 20:21:56 2026
[mdmlot@vbox ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Reflections:

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

What we should look for when deciding the best distribution should almost always be the use case and technical requirements. A Debian distribution would be best for those dabbling in software and cloud development or web hosting while RHEL distributions would work best for enterprise-scale deployments.

2. What are the main differences between Debian and Red Hat Linux distributions?

One main difference between the two distributions is the package manager and software ecosystem of the distribution, with dpkg format and apt manager for Debian, compared to the RPM format and dnf manager of RHEL. Debian based distributions are often 100% community-driven with no commercial fees while many RHEL distributions have licensing fees for their enterprise-scale capabilities. Major versions of Debian distributions often last for 2 years while RHEL distributions often last longer with 3 to 5 years.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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